

Impact of categorization autonomy on effective use and adoption intentions

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ABSTRACT

Category tree view is an omnipresent element in graphical user interfaces where it captures information in terms of a hierarchical structure. These categorization trees facilitate human users' cognitive economy and decision-making. While previous research has investigated the utilities of using unstructured data compared to pre-categorized information by business users, the effectiveness of allowing users the autonomy to create their own categorization hierarchies from generic object types remains unexplored. This paper evaluates the benefits of categorization autonomy in terms of search precision, as an objective measure, as well as subjective intentions to use the system. We examined users' interactions with a platform in information seeking tasks with 201 subjects. Our findings indicate that categorization autonomy leads to superior results, both in terms of effective use and behavioral perceptions. We also found that the impact of categorization autonomy is moderated by task flexibility, such that the benefits are more apparent in tasks that necessitate open-ended search approaches. By focusing on how user-driven categorization influences system interaction, our study contributes to the design of decision support systems that are better aligned with users' cognitive structures and task demands.

1. Introduction

The field of human-computer interaction (HCI) has studied different designs of computer systems and analyzed the interactions between human users and technology with the aim of improving the means of attaining the systems' goals [1]. Research has shown that designs that "match the preferences and information needs of [users] are efficient; those that do not, forgo potential profit" [2, p. 202]. For example, when Intel implemented a verbal advisor to help customers find the relevant software, successful downloads increased by 27 % [2]. Thus, allowing users to choose a design that matches their cognitive style would be advantageous to a singular interface design.

One design choice that determines the efficiency of information assimilation [3] and enables a better cognitive fit with the preferences and needs of users is related to how data is categorized and presented to users. Categories reflect repeating patterns of properties [4], and categorization is the ability to detect recurring characteristics of phenomena [5]. Categories aid humans with the processing of information in a cognitively economical manner [6]. In information systems, category trees are widely used to provide users with the same utilities of abstraction from the way the information in databases is organized [7],

to how items are grouped and presented on almost every user interface [8], whether in Decision Support Systems (DSS), online banking, inventory management systems, or e-commerce. In particular, focusing on improving the decision-making process through the development and evaluation of systems that provide users with enhanced information and support is central to the design of effective decision support systems [9]. From a DSS perspective, categorization tools function as cognitive support mechanisms that assist users in identifying relevant information. Design decisions that shape this structuring – such as granting or restricting autonomy – could therefore impact the effectiveness of decision support.

In this work, we study the effect of *categorization autonomy*, defined as the extent to which users have the ability to define their own categorization schemes from a base level of object types. In order to feasibly examine the impact of categorization autonomy, we chose to focus on e-commerce, since users of such platforms access large amounts of information without the need for prior technical training [10] – making it an appropriate setting to study how users' internal categorization mechanisms can be best utilized in processing information and making accurate decisions [11]. Although other applications such as content management systems or organizational resource planning systems could

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have been fitting contexts to study the benefits of categorization autonomy, we believe that the accessibility offered by e-commerce platforms justifies the choice. While we are not limiting our contributions to the e-commerce context, we acknowledge that pre-defined categorization trees are considered “must-have” design elements in many interfaces [12]; however, not exploring a new design that potentially provides a better fit with the cognitive styles of users is akin to forgoing profit [2]. Furthermore, DSS research has long emphasized the importance of adaptive system features that support decision quality across a variety of user profiles and task contexts [13]. Categorization autonomy aligns with this principle by enabling the system to better fit the needs of individual users.

Prior research on human users' interaction with data categorization includes studies on usage of non-classified data (or instance-based), where only things and their attributes are recorded devoid of pre-defined structures [7]. Namely, it was shown that content generators on a citizen science platform were able to provide higher quality data when they had the option to record their observations free of classification [14,15]. From a different perspective, content consumers in a self-service business intelligence setting were able to write more accurate queries and identify higher quality patterns when they had access to non-classified data versus class-based data [16].

We believe one potential solution that offers individuals the ability to reap the benefits of categorization in decision-making, while also providing a better fit between the categorization scheme and the individual's information needs, is to allow them to define their own categorization hierarchies (rather than stripping data of all categorization/classification, which distinguishes our work from previous efforts). In fact, our premise is that the *most effective* categorization scheme varies substantially across users due to their unique ways of assimilating information [3]. To the best of our knowledge, providing autonomy to general users in creating customized categorizations based on similarity of existing lower-level categories or object types in an inclusive setting, such as e-commerce, has not been studied before. This notion resonates with DSS literature on user-driven modeling and tailoring of information structures [17], which suggests that empowering users to organize information according to their decision logic can lead to better system outcomes.

Rooted in March's [18] learning taxonomy, “two types of system usage can drive individual task performance: exploitation and exploration” [19,p. 236]. Hence, in order to conduct a comprehensive investigation of how categorization autonomy could be used in practice, we follow this taxonomy in our study design. In the context of online searches on an e-commerce platform, exploitation corresponds to searches where users pursue well-defined goals by following a closed and structured approach (e.g., searching for a 16-gigabyte USB storage), whereas exploration involves searches characterized by greater variability and openness in the search process (e.g., browsing for Christmas gifts for friends and family). In this paper, we capture the two modes of system usage with the *Task Flexibility* concept – defined as the extent to which a task allows for multiple ways of achieving the desired outcome. Tasks with low flexibility align with exploitation, where users follow a predetermined and closed-ended path. Conversely, tasks with high flexibility encourage open-ended questioning and variability.

Considering the usage types, we examine the impact of categorization autonomy from an objective usage effectiveness perspective in terms of users' ability to fulfill a certain usage goal, in line with the definition of *effective use* [20], as well as subjective behavioral perceptions. *System Use* has been described in terms of a user, system, and task, and defined a task as a “goal-directed activity” [19, p. 231]. Further, system use and effective use can be distinguished from one another by emphasizing on the performance of “a goal-directed activity to using it in a way that helps attain the relevant goal” [20, p. 633], instead of just being simply able to use the system. With a goal as a “cognitive representation of a desired end-point” [21, p. 491], it is assumed that goal-attainment has objective qualities, which can be operationally

assessed in terms of the performance of an observable behavior [20]. In this paper, we set out to evaluate the efficacy of categorization autonomy from both perspectives of objective performance as well as subjective intentions to use the system given the chance. The objective and subjective analysis is further motivated by the multidimensional nature of Information Systems (IS) success [22]. Demonstrating that categorization autonomy not only helps in the attainment of usage goals (i.e., objective measure), but also leads to higher subjective perceptions, provides strong support for its efficacy as a viable design.

To study the potential benefits of categorization autonomy, we conducted an online laboratory experiment with a total of 201 subjects who were surrogates of actual online shoppers. We developed an experimental website using data scraped from a major North American e-retailer (with annual revenue of more than US\$50 billion) offering subjects the same level of product information as a typical e-commerce platform. Based on random assignment of subjects to three levels of categorization autonomy (i.e., no autonomy as in the source website, partial autonomy, and full autonomy), we found that online shoppers who defined their own product category trees were able to search for products more accurately in both closed and open-ended searches. Furthermore, task flexibility was also found to demonstrate a significant moderating effect, indicating that the effect of categorization autonomy is indeed stronger for tasks with higher flexibility (i.e., exploratory tasks). The behavioral usage intentions were also positively improved for users who had full categorization autonomy. Advantageous results in both objective and subjective metrics for categorization autonomy provide strong evidence for the effectiveness of this design. We also performed a post-hoc analysis to understand the differences between individuals' self-defined categories and their variations from the baseline of the e-retailer's pre-defined scheme. Our findings indicated that individualized categories were indeed dissimilar from one another and the e-retailer's – corroborating that users assimilate information differently [3]. The differences from the baseline were significantly stronger for users who demonstrated greater levels of the openness personality trait, which is associated with the generation of novel ideas and creativity [23].

2. Conceptual development

Categories help human users process information more efficiently, and in information systems, they are used in various settings from database organization [7] to grouping products on e-commerce platforms, aiding shoppers in navigating options. Categorization trees are considered one of the “core elements of a retail e-commerce website” [24,p. 2]. Their importance is highlighted beyond just website search functions, noting that effective searches require knowledge of the domain to input the right keywords [25]. Even with some domain expertise, users might favor browsing categories for a more comprehensive overview, especially on platforms with less effective search engines than Google. In decision support systems, also, taxonomies have been shown as effective decision support aids, particularly for non-expert users [26]. Moreover, even a knowledgeable user still needs good querying skills for efficient searches, as certain terms lead to better results [25] (e.g., “romantic comedy” over “funny movie for a date”). Thus, categorization trees are indeed valuable features for users with average domain familiarity and querying skills. Offering improvements to the categorization tool could increase the utility of this design feature even further.

Prior categorization research in Information Systems has discussed human users' ability to create and interpret categories.¹ The stream that

¹ Automation of categorization is also discussed in the literature (e.g., [27,28]), however, the database design (by humans or machine) is beyond our scope as our focus is on the individual content consumer who uses the categorization scheme for their own individual use.

studies humans' decisions with regards to creating categorization prescribes how basic classes can be identified by a system's designer in order to create better interfaces (e.g., [29]), but that perspective is beyond our scope as it takes the point of view of a designer rather than the end-users. In the other line of research that focuses on human users' interactions with and understanding of categories, we could refer to the instance-based paradigm [7], as a data management approach that is an alternative to class-based. The instance-based paradigm is conceptualized in a two-layered architecture: the first layer includes the instances and their attributes (e.g., a round thing that is red, has a 30-in. circumference, and weighs 620 g), and the other layer includes classes (e.g., basketball, baseball, hockey puck) along with their definitions. This stream developed further, namely, from content-generator [14,15] and content-consumer perspectives [16]. Prior research has found evidence that enabling instance-based data entry for a 'crowd' that uploaded user-generated content to a citizen science platform increased overall accuracy and crowd information quality [14]. The content-consumer perspective investigated users' ability to run queries and also engage in business intelligence and pattern detection on datasets that were free of classification (i.e., instances and their properties). It was found that users of instance-based data achieved higher query accuracy, and performed significantly better in pattern discovery [16].

The present study is different from the instance-based paradigm as we kept the classes assigned to products and thus, did not directly study instances and attributes. Moreover, this paper is not focused on automating the creation of categories within e-commerce, where algorithms certainly play an important role in handling a large number of items that need to be classified. The distinction in this research is in allowing general users to define their categorization trees and hierarchies based on a given set of already specified leaf-level categories (such as television, camera, or fridge).

2.1. Impact of categorization autonomy on effective use

Categorization provides a form of cognitive economy [6] by helping organisms, such as humans, gain the maximum amount of information through minimum effort, optimizing the impact on their finite cognitive resources. This description of categorization plays into what we consider heuristics. Heuristics are problem-solving 'shortcuts' that usually lead to an outcome, which may not be the optimal solution, but use relatively low amounts of cognitive resources to do so [30]. Humans create categories through learning and *information processing*. There are several models that try to explain how information processing occurs in individuals. Here, we discuss the ramifications of self-defined categories from the perspectives of cognitive schemas [3] and decision-making [30].

Cognitive schema theory (CST) posits that human memory is made of 'memory objects' that form 'mental models' [3]. Memory objects are the building blocks that contain general facts about something (e.g., I remember the weather was sunny). CST implies that individuals develop memory objects through life experiences, such as their studies, events, and their culture. Memory objects that are linked to one another form mental models, which tend to be more situational in nature. In the information assimilation process, a model is adjusted to the situation at hand. These mental models allow individuals to form patterns that consist of multiple types of memory objects, such as facts, emotions, and the relations between objects; for example, "I remember the weather was sunny (fact), and I felt good (emotion) – sunny weather makes me feel good (pattern)".

This is an important aspect when looking at pre-defined versus self-defined categorization (i.e., categorization autonomy). According to CST, we understand that an individual's mental models are dependent on their personal experiences. As everyone's personal experiences are unique, we can assume that the patterns they have formed are distinct as well. In this paper's context, these patterns entail the way a product tree is categorized. The tree created by a systems designer in an organization

would have used different memory objects and patterns than the ones a given end-user is familiar with. This could result in an incongruent product tree for the user, leading to inefficiencies in using it.

From a different perspective, Tversky and Kahneman [30] discuss heuristics as mental shortcuts that are used in human decision-making. Initial information cues or starting points also act as heuristics that can influence individuals [31]. In essence, we can consider high-level categories as anchor points for lower-level categories. Tversky and Kahneman [30] discuss that these heuristics function as anchors that bias and limit one's decisions. In the absence of such anchors, an individual tends to have a greater degree of freedom in thinking and decision-making. Hence, we posit that pre-defined categorizations (i.e., the condition with no categorization autonomy) act as anchors that bias one's decisions towards the original designers' perspective. Self-defined categories – once created – could also act as anchors in future use, but they would be more congruent with the given individual's view in the first place (i.e., as the result of full categorization autonomy). Moreover, according to Cognitive Fit Theory [32], performance improves when the information representation aligns with the user's cognitive processes. In other words, allowing users to create their own categorization schemes provides a better cognitive fit, which should lead to higher search precision, as users can structure information in a way that mirrors their internal problem-solving strategies. Recent DSS studies have extended Cognitive Fit Theory by showing how the alignment between information representation and task characteristics improves decision performance across various contexts. For example, the literature has demonstrated that presentation format interacts with product type to shape decision quality in e-commerce, highlighting the importance of matching system features with the nature of the user's task [33]. Similarly, it was also shown that even within data visualization contexts, users experience varying levels of cognitive effort depending on how well the design fits the task demands [34]. These perspectives emphasize the need for systems to support diverse cognitive processes.

Based on the aforementioned perspectives, our proposition is that users who define their own categories will assimilate information better, and hence, will be able to attain their goals of using the system. A goal can be conceptualized as a "cognitive representation for a desired end-point" [21, p. 491]. Building on this, effective use was defined as "using a system in a way that helps attain the goals for using the system" [20, p. 633]. The goal-attainment factor is what distinguishes effective use from general use. Further, assuming objective qualities to goal-attainment, it was posited that effective use can operationally be assessed in terms of performance of an observable behavior [20]. In e-commerce, the goal is to search and find products, and its fulfillment can be measured in terms of users being able to correctly search and identify items with certain attributes (or reach the 'desired end-point' in terms of a goal). Therefore, we hypothesize that because of better assimilation of information, subjects having autonomy to create their own categorization schemes will be more successful in searching and identifying correct items (and achieve their usage goals):

H1. : *Categorization autonomy leads to higher levels of search precision while using the system.*

Search precision is the operationalization of *effective use*, which we calculated as the ratio of the correctly identified items over the number of pre-specified items (i.e., the goal) in this study.

2.2. Task flexibility moderating the impact of categorization autonomy on effective use

While categorization autonomy offers users a greater degree of control over how they structure information, its effectiveness may depend on the nature of the task being performed. In particular, *Task Flexibility* - defined as the extent to which a task allows for multiple ways of achieving the desired outcome - may shape how beneficial categorization autonomy is in improving search precision and decision-making.

Some tasks are inherently flexible and allow users to pursue various paths and options (e.g., browsing for a birthday gift for a friend), while others are relatively rigid and afford limited flexibility (e.g., locating a replacement cartridge for a particular printer model). This variation in how a task can be approached may influence whether users benefit more from a predefined categorization tree or from the ability to define their own categories.

This distinction aligns with Cognitive Fit Theory [32], which posits that task performance improves when the representation of information matches the cognitive processes required for the task. When a task is flexible, users can benefit more from the ability to define categories in a way that reflects their personalized strategies and evolving sense of relevance. Categorization autonomy, in this context, provides a better cognitive fit by supporting adaptive information structures. In contrast, when tasks are inflexible and allow only a narrow set of procedural steps or criteria, the user may not require (or be able to take full advantage of) the freedom offered by categorization autonomy, as the rigid nature of the task already constrains their decision-making path.

A related lens is provided by Task-Technology Fit Theory [35], which suggests that a system is more likely to enhance performance when its functionality matches the demands of the task. In flexible tasks, categorization autonomy increases the alignment between task and technology by enabling users to construct a structure that suits their evolving needs. For inflexible tasks, however, the fit between technology and task may not be as much improved by the introduction of autonomy, as the standard categorization approach may already be predefined by the task's limited degrees of freedom. Based on this rationale, we expect that propose the following hypothesis:

H2. : *Task flexibility moderates the relationship between categorization autonomy and search precision such that the effect of categorization autonomy would stronger for more flexible tasks.*

2.3. Effect of categorization autonomy on behavioral perceptions

As mentioned earlier, when assessing the efficacy of a design, in addition to evaluating objective performance measures (such as search precision), understanding behavioral dimensions is also crucial in determining its success and adoption by users [22]. Providing online users with the autonomy to create self-defined categorization schemes could affect their intentions to use a platform. We study the relationship between categorization autonomy and usage intentions through the mediation effect of perceived restrictiveness of the interface. Silver [36] defines the restrictiveness of a system as the extent to which a system constrains the user's "decision-making processes to a particular subset of all possible processes" [36, p. 52]. Restrictiveness is considered a relative term, where the set of supported processes or actions in a system could be larger or more limited compared to another system's set of supported actions [36]. In our context, perceived restrictiveness is relevant since in the fixed categorization condition, the system does not allow the user to perform any actions with regard to categorization of concepts – hence, that condition is restricted. With categorization autonomy, on the other hand, the system allows users to create categorization trees based on their world view. As a result, categorization autonomy is less restrictive and is expected to be perceived as such by the users.

We predict that subjects would have lower intentions to use an interface that has fixed categorization schemes compared to a system that offers categorization autonomy. This reasoning is in line with prior work [37] in which they studied decision support aids with additive-compensatory, eliminatory, and hybrid strategies and showed that when a user's desired support functionalities were not offered by a decision support system, the psychological reactance reduced their intentions to use the system - as per the reactance theory [38]. In their study, the relationship between decision strategy and adoption intentions was mediated by perceived restrictiveness (resulting from

different decisions). From a different angle, it has been discussed how the "autonomy over the communication process" [39, p. 737] with a system would lead to higher adoption intentions. In our context, we expect categorization autonomy to positively affect usage intentions, and this relationship to be mediated by perceptions of restrictiveness:

H3. : *Categorization autonomy leads to higher usage intentions.*

H4. : *Perceived restrictiveness will mediate the relationship between categorization autonomy and usage intentions.*

The proposed nomological research model is captured in Fig. 1.

While we recognize that in a longitudinal study of habits, perceptions could affect performance, or vice versa, we considered the objective performance to be independent from subjective perceptions during our study (which was a one-time task). We believe that in some situations, an individual may not necessarily have a positive perception towards a task (e.g., not enjoying a warm-up drill before a sports game), yet, positive results may be achieved from an objective perspective (e.g., better agility while playing the game). Or, alternatively, subjects may have positive perceptions (e.g., enjoying candies before a match), and as a result, have poorer objective performance (e.g., experiencing sugar crash and fatigue). Hence, we decided to investigate objective and subjective measures independently of one another and did not make any hypotheses about their possible interactions.

As mentioned, to thoroughly inspect the phenomenon, we conducted an experiment that operationalized March's [18] exploitation-exploration taxonomy as task flexibility, moderating the impact of categorization autonomy on effective use. The IS literature [20] also adapted March's taxonomy and defined exploitation of information as a task involving the use and search for things already known or a routine execution of knowledge in a closed-ended task. Exploration of information is the search for things in an open-ended setting.

3. Methodology and experimental setup

The concept of 'effective use' – as the users' ability to fulfill the goals of using a system – is often operationalized by performance [20]. To study this in the context of categorization autonomy, we created a website to simulate the task of online shopping. Participants' ability to find products that meet their search goals is a measure of performance or effective use.

We hired an experienced developer to create the experiment's website by scraping the contents of a prominent North American Multinational Consumer Electronics (NAMCE) retailer with an annual revenue of more than US\$50 billion. The scraped product data was imported into our experimental website, and using the WooCommerce – Store Exporter plugin, a list of category names and associated URLs was generated in JSON format to populate the interactive platform. The experimental website included over 10,000 products, with prices, descriptions, and images, and was fully functional – allowing users to browse, add items to their cart, and proceed to a checkout screen.

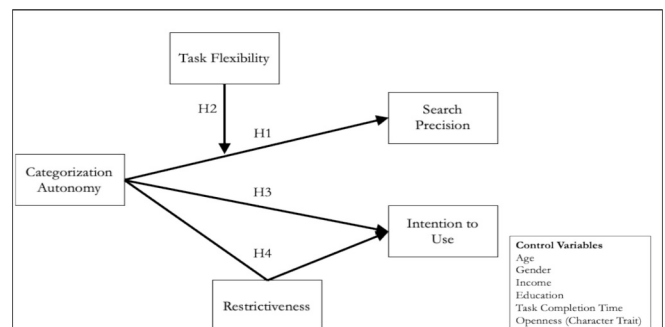


Fig. 1. Nomological research model.

Separate environments were created for each condition, all containing the same products to ensure informational equivalence [40]. The product category tree on NAMCE was captured in a three-level hierarchy; for example, Appliances as a top level, included Major Appliances and Small Kitchen Appliances in the middle tier, and then general product types such as Dishwashers and Coffee Makers as leaf-level general product types. We also adopted this three-level categorization hierarchy in our study – Fig. 2 shows a screenshot of our website.

The NAMCE contained around 250 leaf-level categories (e.g., tablet, air conditioner, laser printer). For practical reasons, we had to limit the number of leaf-level categories as it would have taken a considerable amount of time to classify the whole set of 250 categories. In the real world, in case categorization autonomy was offered by a platform, users could create their categories over multiple sessions; however, that would not have been feasible for our studies. Hence, we conducted multiple trials online with 10 Amazon Mechanical Turk (AMT) workers in each iteration, and trimmed 30 to 40 leaf-level categories (while maintaining a relatively consistent distribution at higher levels of the tree) in each attempt until the categorization time reached the average of 15 min with 78 leaf-level categories remaining. Given that attention typically declines after 10–15 min [41], we set 15 min as a reasonable time for participants to create categories before starting the main task.

We used three conditions in our studies. The first condition, or control, was based on the NAMCE's website and all three levels of the category tree were fixed (i.e., no autonomy in changing the categories by users) – from here on we use *fixed categorization* and no autonomy interchangeably. Our treatment condition provided *full autonomy*, and showed subjects only the leaf-level categories. Their interface had a feature that allowed them to create high-level (i.e., Level 1) and subcategories (i.e., Level 2) and populate them by dragging and dropping the leaf-level categories (i.e., Level 3) inside them. The flexible categorization interface was developed using JavaScript on the Vue.js framework, and included an 'edit' interactive panel. Fig. 3 shows the edit page where a subject created one top-level category named "Home" with two subcategories of "Kitchen" and "Living Room". The leaf-level categories on the right-side pane could be dragged and dropped under them by the users. The users' self-defined categories were stored as structured JSON objects.

We also created a middle condition where Level 1 and Level 2 were the same as the source website, but participants had to manually drag and drop the leaf-level categories under these fixed levels. We consider this condition to offer *partial autonomy*. We did not develop formal hypotheses regarding this condition, as it is not part of our theoretical development. Rather, we included this to rule out alternative explanations for the effects observed in the full autonomy condition. Namely, the partial autonomy condition helps clarify whether the observed improvements in performance and perceptions were due to the effect of categorization autonomy itself, or from greater time spent engaging with the interface. The partial autonomy condition strengthened the internal validity of our findings by serving as a control for potential confounds related to effort and user involvement.

3.1. Experimental design

We adopted a 3×2 between-subjects experimental design to investigate the impact of Categorization Autonomy (fixed, partial, full) on users' search precision, and to examine whether this relationship is moderated by Task Flexibility (low in exploitive search vs. high in exploratory search). Categorization Autonomy was the primary manipulated factor, representing different levels of user control over the categorization structure. Task Flexibility, treated as a moderating factor, is conceptualized as the degree of variability in how a task can be approached and completed. Higher task flexibility enables greater variability and openness in decision-making processes, whereas lower task flexibility constrains users to follow a more closed and predefined path. Participants were randomly assigned to one of the six experimental

conditions resulting from the factorial combination of the two variables. This design enabled us to assess the main effect of Categorization Autonomy on search performance, as well as whether the strength of this effect varied across levels of Task Flexibility. After reviewing the consent form, subjects received pre-experiment questions, which collected information about e-commerce experience, openness (a big-five personality trait [42]), and demographics.

Subjects in all six conditions viewed a training video that demonstrated browsing, ordering, and adding to cart functions. Subjects in the partial and full categorization autonomy groups received an additional five-minute tutorial on using the 'edit categories' page and creating their own categories – as the category 'edit' feature was only available to them. We used different products in the training demonstration (which were not available in the main experiment) to avoid biasing the participants.

Next, the subjects in the partial and full autonomy conditions (whether in high task flexibility or low) were asked to edit or create their categories respectively, and the fixed categorization (i.e., control) group was asked to spend five minutes inspecting the categories. The mandatory inspection time given to the control group allowed us to provide an equivalent condition (with regard to familiarization and engagement) for browsing the categories between all three conditions. Further, the time that the participants spent on the page was measured to account for any differences between the groups. These steps were consistent in all conditions, and as reported in the post-hoc analysis section, the time spent in this stage was not a significant covariate on the results. Then, subjects received the scenarios for the respective studies and were asked to perform the assigned task. At the end, participants answered survey questions regarding their usage intentions (Appendix A).

3.2. Sample selection

For our studies, we used Amazon Mechanical Turk (AMT) workers belonging to the 'master worker' category – top performers who were granted the 'master' qualification by Amazon – with more than 100 jobs completed and at least a 95 % approval rate. We also limited the sample to participants located in the United States or Canada (where the source e-retailer also operates). Several studies have shown that AMT workers exhibit comparable biases and performance to other participants in traditional laboratory experiments [14,43]. In addition, we believe that AMT subjects are appropriate surrogates for typical e-commerce users with similar demographic profiles; hence, their recruitment is justified. Based on the initial statistical power analysis on a pilot with 20 subjects, we decided on a sample size of around 35 in each condition based on the desired power of 0.8. Moreover, it has been stated that for a sample containing minor subgroups, a minimum of 20 to 50 elements is needed per group [44]. The subjects received fixed compensation depending on the experimental condition that was higher than the minimum hourly wage in the United States.² Participants were 45 % female and 55 % male with 36 % in the age bracket of 30–40 years old. We eliminated duplicate entries, treatment participants who had not created a category, subjects who had not placed an order, and those who failed the trap question in the survey.³

The scales for the behavioral measures were tested for validity and reliability (Appendix B). All the variables exhibited high factor loadings and thus warranted no further adjustments. Next, we examined the Cronbach's alpha and Average Variance Extracted (AVE) measures for

² In pilot studies, control subjects averaged 30 min, while partial and full autonomy groups averaged 37 and 45 min, respectively. Accordingly, participants were paid \$6 (fixed), \$7 (partial autonomy), and \$8 (full autonomy).

³ We initially recruited 77 subjects in the fixed, 73 in the partial, and 75 in full categorization autonomy conditions. After the exclusions, we analyzed 69 subjects in fixed, 67 in partial, and 65 in the full categorization autonomy group.

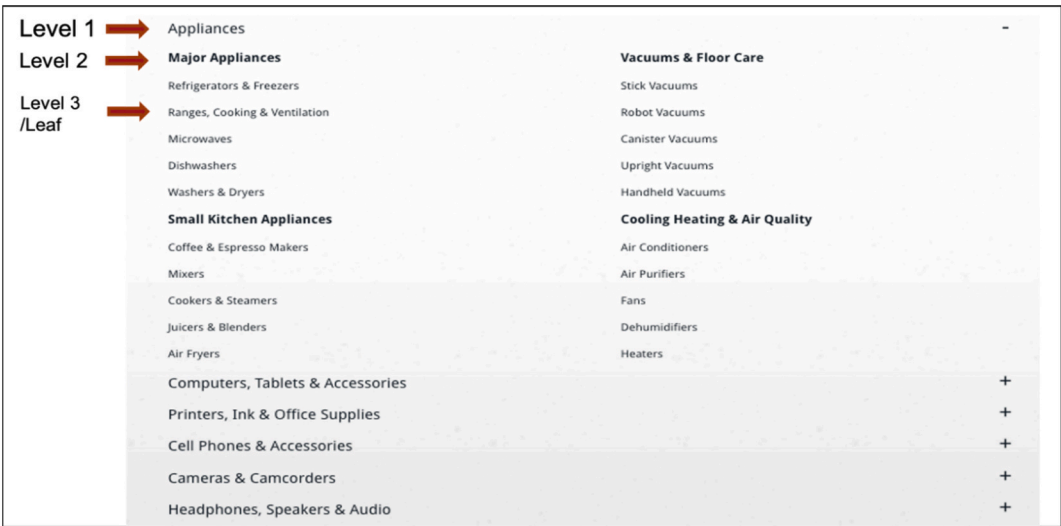


Fig. 2. Three levels in the product category tree, as in the source website.

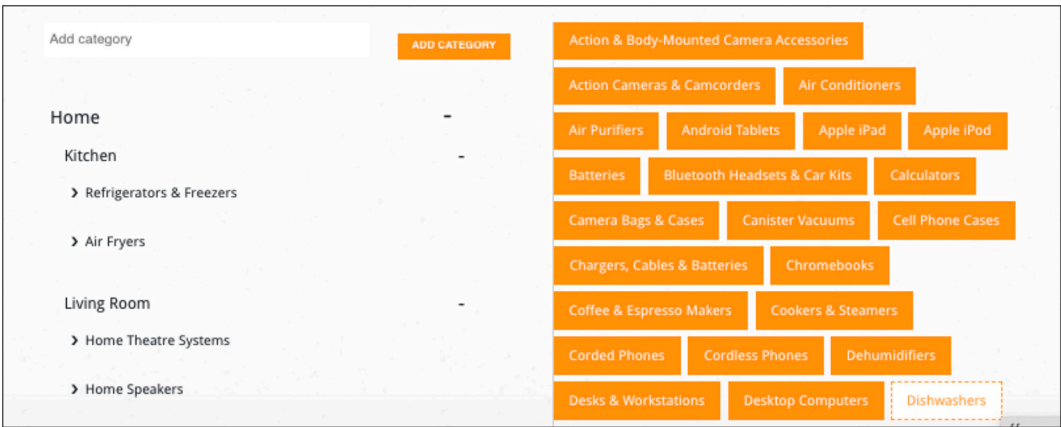


Fig. 3. Treatment Condition Interface with Category Edit Feature.

the different constructs. The results demonstrated a Cronbach's alpha of above 0.9 for all constructs, which indicated high reliability [45]. The AVE of all constructs was high as well (>0.5), indicating high convergent validity [46]. Appendix B provides the details.

3.3. Experiment

As mentioned, subjects were randomly assigned to one of the main conditions of control (i.e., fixed categorization), partial categorization autonomy with the first two levels of the category tree fixed according to NAMCE's design, and treatment conditions (i.e., full categorization autonomy), and either performed the closed-ended exploitive task (i.e., low task flexibility), or the open-ended exploratory search (i.e., high task flexibility). The subjects in the fixed categorization group were instructed to study the product tree, while the subjects in the partial and full categorization autonomy conditions had to categorize 78 leaf-level product types into higher level categories. We employed a live chat function for subjects in every condition, and we were reachable to them throughout the experiment. The actual task scenario was locked behind a password-protected page. Subjects in the fixed condition received the password after five minutes, and subjects in the partial and full autonomy condition had to let us know when they were done with the categorization activity, and once we verified their completion of building the category trees, we offered them the password that allowed them to proceed to the main task. The scenario for the tasks is available in

Appendix C.

Subjects did not have access to a keyword search function and had to locate the items using the categorization tree of their respective conditions. The search function feature was removed due to the intention to run a controlled laboratory experiment, where extraneous factors and their potential interactions were held constant to isolate the effect of categorization autonomy. Once subjects finished adding the items to the shopping cart, they proceeded to the post-experiment questionnaire about behavioral perceptions. Table 1 presents the descriptive statistics.

The first hypothesis was related to subjects' ability to search and identify the correct products based on the given scenario. Two independent coders went through the shopping carts of subjects in all three conditions and rated whether the item was acceptable with regard to the specifications. The intra-class correlation between the two coders was 95 %, which indicates a high degree of reliability.

Subjects received a point for every correct item they had identified (e.g., one point if they had chosen a 27-in. monitor as per the scenario, or an air-fryer as a gift for a cooking enthusiast, but no points were awarded for a product that did not match the specifications). To compare the precision of the results, we calculated the ratio of correct answers to the total number of required items (i.e., dividing each score by the number of items subjects were instructed to buy). Since categorization autonomy can be treated as an ordinal variable (fixed categorization or no autonomy $<$ partial autonomy $<$ full autonomy), we ran regression tests (after checking for normality of distribution) to

Table 1
Descriptive statistics.

Variable Condition	Search Precision Mean (SD)	Perceived Restrictiveness Mean (SD)	Intention to Use Mean (SD)
Fixed – Low Flexibility, <i>N</i> = 36	0.822 (0.233)	4.231 (1.441)	4.389 (1.573)
Fixed – High Flexibility, <i>N</i> = 33	0.737 (0.371)	4.689 (1.592)	4.263 (1.659)
Partial autonomy – Low Flexibility, <i>N</i> = 30	0.811 (0.226)	3.533 (1.464)	5.577 (1.229)
Partial autonomy – High Flexibility, <i>N</i> = 37	0.677 (0.329)	3.387 (1.827)	5.811 (1.158)
Full autonomy – Low Flexibility, <i>N</i> = 34	0.971 (0.068)	3.069 (1.618)	4.932 (1.395)
Full autonomy, High Flexibility, <i>N</i> = 31	0.914 (0.171)	2.785 (1.447)	5.204 (1.497)

investigate which group orders showed the effect. The ordinal analysis is in line with a prior study with a similar design [47] on decomposition of conceptual models and the adherence to Good Decomposition Principles at three levels (comparing good vs. bad, good vs. moderate, and moderate vs. bad). Similar to their analysis, we investigated whether higher degrees of categorization autonomy led to advantageous results. Table 2 shows that offering categorization autonomy resulted in an overall increase in search precision (0.077, $p < 0.05$; Model 1). In our group-wise comparisons, we find that search precision for subjects who had full categorization autonomy was significantly higher (0.155, $p < 0.01$; Model 2) than users of pre-defined categories from the original retailer's website (i.e., fixed categorization or no autonomy). Subjects in the full autonomy condition also significantly outperformed subjects in the partial autonomy (0.204, $p < 0.01$; Model 3), but the difference in performance between fixed categorization and partial autonomy conditions was not statistically significant. This result is expected, as only the full autonomy allowed participants to adjust the information structure to fit their cognitive processes [32], whereas partial autonomy constrained them to fixed Level 1 and 2 categories, limiting flexibility to product placement only at the leaf level. Overall, the first hypothesis was corroborated.

We tested the moderation effect of Task Flexibility (exploitive or closed-ended vs. exploratory or open-ended tasks) on the relationship

Table 2
Regression analysis for impact of categorization autonomy on search precision – Hypothesis 1.

Outcome: Search Precision	Model 1 Coefficient (Standard Error)	Model 2 Coefficient (Standard Error)	Model 3 Coefficient (Standard Error)	Model 4 Coefficient (Standard Error)
Categorization Autonomy (0 = Fixed; 1 = Partial; 2 = Full)	0.077 (0.019)**			
Fixed vs. Full Autonomy (0 = Fixed; 1 = Full)		0.155 (0.039)**		
Partial v. Full Autonomy (0 = Partial; 1 = Full)			0.204 (0.327)**	
Fixed vs. Partial Autonomy (0 = Fixed; 1 = Partial)				−0.048 (0.051)
<i>N</i>	201	134	132	134
<i>R</i> -Sq	0.0570	0.1006	0.1729	0.0067

** $p < 0.01$ * $p < 0.05$.

between categorization autonomy and search precision as our second hypothesis. The results in Table 3 show that the interaction effect of task flexibility and categorization autonomy is significant in a continuous operationalization (0.179, $p < 0.01$; Model 1). After checking for normality of the distribution, we found that task flexibility moderates the effect of categorization autonomy on search precision in comparison of fixed vs. full autonomy (0.161, $p < 0.05$; Model 2), and also in comparison of partial vs. full autonomy (0.229, $p < 0.01$; Model 3).

To examine the influence of categorization autonomy on the usage intention (Hypothesis 3), we used regression models with robust standard errors, as shown in Table 4. Similar to the previous analyses, we first ran regression analysis using the continuous operationalization of categorization autonomy to check if it has a significant effect on intentions to use. Once we confirmed that there was a significant effect of categorization autonomy on intentions to use (0.362, $p < 0.01$; Model 1), we then ran regression tests (after checking for normality of distribution) to investigate which group orders showed the effect. From our analysis of the different subgroups (Models 2–4), H3 is corroborated in the comparison between the fixed vs. full categorization autonomy (0.359, $p < 0.01$; Model 2) and fixed vs. partial categorization autonomy (1.373, $p < 0.01$; Model 4).

Next, to study the mediating role of restrictiveness in the relationship between categorization autonomy and usage intentions (Hypothesis 4), we tested both the product of coefficients (Sobel test) and the bootstrapped confidence interval of the indirect effects (following Preacher and Hayes [48]). In the Sobel test, we checked if the indirect effect was significantly different from zero. The Sobel test assumes normality in the distribution of the indirect effect. Methodologists therefore recommend that it be supplemented with bootstrap confidence intervals, which do not make assumptions about the shape of the sampling distribution [48]. If the confidence intervals exclude zero, the indirect effect (i.e., mediation) is considered meaningful. We therefore calculated bias-corrected and accelerated confidence intervals across 5000 bootstrap resamples.

Table 5 provides the coefficients and standard errors of the indirect, direct, and total effects of the mediated relationship between categorization autonomy and usage intentions. The mediation analysis provides support for H4 for the two comparison groups fixed vs. full autonomy (0.482, $p < 0.01$, Sobel Test, 0.481 $p < 0.01$ Preacher and Hayes Test) and fixed vs. partial categorization autonomy (0.516, $p < 0.01$, Sobel Test and Preacher and Hayes Test). In the mediation Table 4, the 'a

Table 3
Regression analysis for the moderation effect of task flexibility – Hypotheses 2.

Interaction Effects Outcome: Search Precision	Model 1 Coefficient (Standard Error)	Model 2 Coefficient (Standard Error)	Model 3 Coefficient (Standard Error)	Model 4 Coefficient (Standard Error)
Task Flexibility	−0.179 (0.059)**	−0.152 (0.066)*	−0.223 (0.058)**	−0.072 (0.068)
Categorization Autonomy (0 = Fixed; 1 = Partial; 2 = Full) * Task Specificity	0.079 (0.035)*			
Fixed vs. Full Autonomy (0 = Fixed; 1 = Full) * Task Openness		0.161 (0.069)*		
Partial v. Full Autonomy (0 = Partial; 1 = Full) * Task Openness			0.229 (0.062)**	
Fixed vs. Partial Autonomy (0 = Fixed; 1 = Partial) * Task Openness				−0.067 (0.082)
<i>N</i>	201	134	132	134
<i>R</i> -Sq	0.0640	0.0734	0.1655	0.0394

Table 4
Regression analysis for the effect on usage intentions – Hypothesis 3.

Outcome: Intention to Use	Model 1 Coefficient (Standard Error)	Model 2 Coefficient (Standard Error)	Model 3 Coefficient (Standard Error)	Model 4 Coefficient (Standard Error)
Categorization Autonomy (0 = Fixed; 1 = Partial; 2 = Full)	0.362 (0.131)**			
Fixed vs. Full Autonomy (0 = Fixed; 1 = Full)		0.359 (0.128)**		
Partial v. Full Autonomy (0 = Partial; 1 = Full)			−0.653 (0.225)	
Fixed vs. Partial Autonomy (0 = Fixed; 1 = Partial)				1.373 (0.239)**
N	201	134	132	134
R-Sq	0.0383	0.0532	0.0580	0.1926

** $p < 0.01$ * $p < 0.05$.

Table 5
Analysis on the Mediation Effect of Restrictiveness on usage Intentions – Hypotheses 4.

Outcome: Intention to Use	a Coefficient	b Coefficient	Indirect Effect aXb	Direct Effect c'	Total Effect c	Sobel Test	Preacher & Hayes
Categorization Autonomy (0 = Fixed; 1 = Partial; 2 = Full)	−0.698 (0.134) **	−0.561 (0.053) **	0.391 (0.084)**	−0.029 (0.108)	0.362 (0.126)**	0.391 (0.084)**	0.391 (0.091)**
Fixed vs. Full Autonomy	−0.697 (0.131) **	−0.691 (0.060) **	0.482 (0.099)**	−0.122 (0.101)	0.360 (0.129)**	0.482 (0.099)**	0.481 (0.100)**
Partial v. Full Autonomy	−0.424 (0.277)	−0.405 (0.062)**	0.172 (0.115)	−0.826 (0.200)**	−0.654 (0.227)	0.172 (0.115)	0.171 (0.114)
Fixed vs. Partial Autonomy	−0.970 (0.272) **	−0.532 (0.061) **	0.516 (0.156)**	0.857 (0.203) **	1.373 (0.241) **	0.516 (0.156)**	0.516 (0.171)**

** $p < 0.01$ * $p < 0.05$.

coefficient' represents the effect of categorization autonomy on restrictiveness and the 'b coefficient' represents the effect of restrictiveness and intention to use. We note that the effect of categorization autonomy is negative and significant for the fixed vs. full autonomy (−0.697, $p < 0.01$) and the fixed vs. partial autonomy (−0.970, $p < 0.01$) comparisons. Furthermore, we find a negative and significant relationship between restrictiveness and intentions to use across all our analyses (b coefficient). Lastly, in the case of the fixed categorization vs. full autonomy conditions, we observe a full mediation as the direct effect becomes insignificant in the presence of the mediator. For the fixed vs. partial categorization autonomy comparison, however, we observed a partial mediation.

Overall, we believe the benefits of categorization autonomy would be best realized when the task itself is free of prior anchoring biases (discussed earlier, referring to [3]). Considering that exploratory tasks have more open-ended requirements, categorization autonomy provided a more open and flexible means of navigation for users; hence, the moderating effect of Task Flexibility. Moreover, categorization autonomy positively impacts usage intentions, particularly when users are allowed to explore the categories and select items according to their own knowledge and expectations with greater freedom. A more elaborate behavioral research model regarding additional antecedents to adoption is presented in Appendix D.

4. Post-hoc analyses

4.1. Heterogeneous treatment effect

The overarching premise of this paper is that the individuals' mental models depend on their personal experiences, which results in significant differences in how they categorize items. To test this premise and gain a deeper understanding of the differences between personalized categories created by participants in the treatment group (where they had categorization autonomy, and as a result, one could observe the full effects), we captured the vector representations of the product trees of each participant's categorization scheme. Using vector representations [49,50], one can compare the similarities and differences between participants in the treatment group and investigate the antecedents to their differing product trees. The vector representations for two hypothetical participants with five products are illustrated in Fig. 4.

In our analysis, the full vector representation is generated using all combinations of category labels created by the 65 treatment participants⁴ and the NAMCE's scheme (as a baseline for comparison). Adopting such a high-dimensional representation (each new combination of categories creates a new dimension) of the categorization trees is possible in our sample because (a) our experiment was restricted to

creating a categorization tree of a maximum of three levels (e.g., Appliances - > Large appliances - > Dishwashers – similar to NAMCE's design) and (b) the number of leaf categories was limited to 78 items. To further reduce the number of possible combinations of categories, one of the authors analyzed the participant-created category names for synonyms, and another evaluated the identified synonyms to ensure validity. During this exercise, category names that were found to mean the same concept (e.g., computer accessories and peripherals, or cell phone and mobile) were identified as equivalents. Our final representation was generated by evaluating all combinations of categories created by the participants in the treatment group, and the NAMCE category tree used for the control group comprised of 247 (category combinations; rows in Fig. 4) and 78 (leaf categories or products, columns in Fig. 4).

4.1.1. Analysis of the participants' product category trees

The vector representation of the product category tree offers the advantage of quantifying and comparing the similarities (or rather, dissimilarities) of category trees created by the participants in treatment conditions. Using vector representations, we studied how participants differ from each other by calculating the cosine similarities of the participants' category trees. We note that the participants' category trees are considerably different from one another, with a maximum cosine

⁴ We combined treatment subjects from both levels of task flexibility, since the task was presented after the categorization trees were created, and the scenario had no effect on subjects at the creation stage.

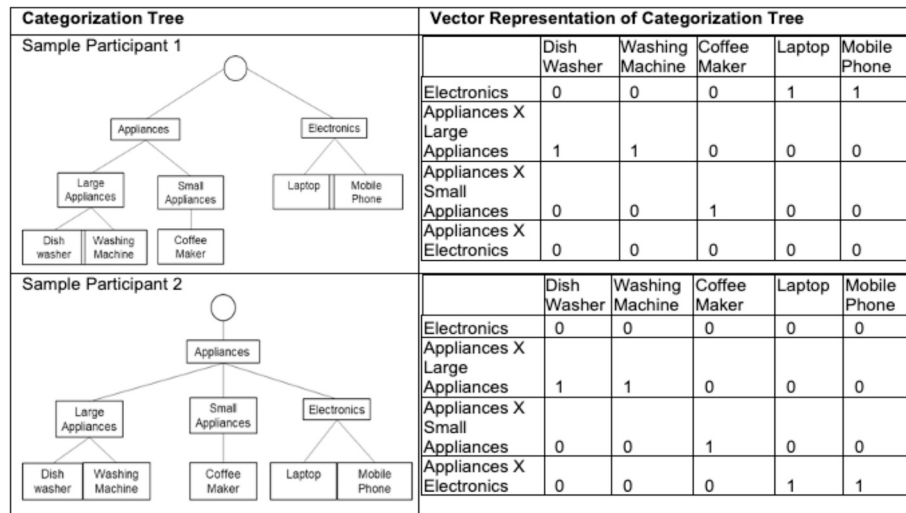


Fig. 4. Vector Representation of Product Categorization Tree.

similarity of 0.58 between them. Further, we find that the participants' self-defined category trees differ considerably from the baseline NAMCE category tree, having a mean cosine similarity of 0.36. The findings support the premise that individuals tend to develop unique schematic representations based on prior experiences [3,31].

4.1.2. Antecedents to the differing expressions of the product category trees

Given the differences reported earlier, we investigated whether demographic and personality traits of participants might have an influence on the cosine similarity measures of treatment users' vector representations of self-defined categories and NAMCE's schema. As a major antecedent of the diversity of expression of individuals, literature points to the important role that openness to experience plays in divergent thinking [51,52]. Specifically, openness is one of the key personality traits that is necessary to produce new ideas [23]. In our experiment, openness was operationalized as the extent to which a person was open to experiences and their degree of creativity [42]. Along with openness, we also collected demographic information, including age, gender, income, and level of education. Table 6 provides the coefficients and standard errors in estimating the cosine similarity of the participant's category tree with the NAMCE category tree using the Ordinary Least Squares (OLS) regression model. Based on the analysis, we found that the openness factor was negative and significant ($-0.31, p < 0.05$; Table 6). This confirms prior literature suggesting that openness is related to how diverse individual expressions are. In other words, this implies that users with higher levels of openness (as a personality trait), tend to create product categorization trees that are less similar to a pre-defined scheme created by the vendors. The other demographic factors did not have a significant effect on the dissimilarity measure of self-defined categories to the baseline.

Table 6

Demographics and (Dis)similarity with NAMCE as a baseline.

Dependent Variable: Cosine Similarity to NAMCE	Regression Coefficients (and Standard Errors)
Openness	-0.31^* (0.14)
Age	-0.147 (0.791)
Gender	0.560 (2.712)
Income	-0.656 (0.935)
Level of education	-0.258 (2.03)
R-Sq	0.103
N	65

** $p < 0.01$ * $p < 0.05$.

4.2. Impact of time

We analyzed the effect of time spent on observing pre-defined categories, or creating self-defined categories (i.e., before the scenario was given) in both conditions of task flexibility as covariates on overall search precision. The time spent on observing or creating categories (depending on condition) in the exploitive scenario had the p -value of 0.984. Similarly, in the exploratory setting, the time spent on observing or creating categories had the p -value of 0.328 as a covariate on search precision. We conclude that the time spent before the tasks did not have a significant effect on performance. We also checked for the effect of time spent on creating or browsing categories (i.e., pre-task time) and the time spent on the task as covariates on usage intentions, and neither were significant ($p = 0.483$ and $p = 0.363$ respectively).

Comparing the actual time spent on the experimental task (post-initial observation or creation of categories depending on condition), we found significant differences in both task types with the order of longest time spent by subjects in the partial categorization autonomy condition, then by the treatment (full categorization autonomy), and control (no autonomy). Task completion time could be considered a measure of efficiency; however, since subjects were not instructed to finish the task as soon as possible, we do not necessarily consider the observed difference to indicate inefficiency.

5. Discussions and conclusion

We studied categorization autonomy as a design principle and explored its impact in terms of effective use, as well as usage intentions. While the primary focus and therefore the contribution of our research lies within the domains of Human-Computer Interaction (HCI) and Decision Support Systems (DSS), it also aligns with the components of a design theoretical model [53], as it defines a clear purpose – enhancing user decision-making through categorization autonomy – supported by theoretical constructs such as task flexibility and effective use. The principles of form and function are reflected in hierarchical categorization trees, while the evaluation of performance and behavioral outcomes provides testable propositions. The design is grounded in knowledge drawn from cognitive schema, cognitive fit, task-technology fit, and reactance theories.

Our study was set in the context of e-commerce, as the users access and browse large amounts of information, making it an appropriate platform [10]. The benefits of categorization autonomy can be realized in other contexts and business domains as well. For example, YNAB (You Need a Budget), one of the most popular budgeting applications, offers

users the autonomy to create and customize budget categories that align with their personal financial goals, enabling a tailored expense management approach.⁵ This autonomy has generated user engagement, as evidenced by the r/YNAB subreddit, where thousands of members share their unique categorization strategies.⁶

Our findings provide strong empirical support for the positive impact of categorization autonomy on users' search precision when compared to fixed categorization. Furthermore, the moderating effect of Task Flexibility was statistically significant, indicating that the benefits of categorization autonomy are more pronounced in tasks characterized by higher flexibility. We also included a mid-level condition with partial categorization autonomy, in order to rule out alternative explanations of learning effect (improving effective use) and extended interaction with an interface leading to more favorable perceptions (e.g., intention to use). Our results were consistent in that treatment (i.e., full categorization autonomy) performed better than both fixed categorization and partial autonomy condition in terms of search precision. However, the difference between fixed and partial autonomy conditions was not statistically significant. This could indicate that offering partial autonomy at the lower levels (product level), may not provide the full benefits of categorization autonomy, since the top two category levels (from the source) were not congruent with the users' mental models.

We also examined the effect of categorization autonomy on usage intentions, as well as the mediation effect of perceived restrictiveness on this relation. In the exploitive searches, we found the direct relationship of categorization autonomy on usage intentions (H3) to be statistically significant, so was the mediation effect of perceived restrictiveness (H4), particularly in the comparison between control and treatment (i.e., no categorization autonomy vs. full autonomy). In comparing the partial autonomy condition to full categorization autonomy, hypotheses H3 and H4 were not corroborated. These results were consistent in both exploitive and exploratory searches. The summary of our results that were corroborated or rejected are presented in Table 7. We conclude that categorization autonomy leads to greater search precision (i.e., objective effective use) compared to conditions with partial or no categorization autonomy. The impact of categorization autonomy on usage intentions is significant, particularly in comparison between fixed vs. full autonomy, or fixed vs. partial autonomy.

5.1. Implications

This paper contributes to categorization, schema theory, and human-computer interaction literature. Categorization research has mainly

Table 7
Summary of results – corroborated and rejected hypotheses.

Hypothesis	H1	H2	H3	H4
Condition	Search Precision	Moderation by Task Flexibility	Intention to Use	Mediation by restrictiveness
Fixed Categorization vs. Full Autonomy	Yes	Yes	Yes	Yes
Partial v. Full Categorization Autonomy	Yes	Yes	No	No
Fixed Categorization vs. Partial Autonomy	No	No	Yes	Yes

focused on creating categories that provide higher utility and cognitive economy in general applications. Our findings indicate that greater utility can be achieved in terms of higher search precision when users create their custom-made categorization trees (i.e., the “what” and “how”). It is also in line with schema theory [3], which indicates that humans assimilate information better when they create mental models that are congruent with their experiences (i.e., the “why”) and enhance their decision-making [54]. The post-hoc analysis illustrated this distinction; specifically, the cosine similarities of the participants' self-defined categories were divergent from one another and the baseline (i.e., for “whom”). This difference could partly be attributed to the openness character trait, which was a statistically significant determinant of the dissimilarity of the individuals' vector representation matrices from the NAMCE baseline.

From the human-computer interaction perspectives, we investigated a novel data representation design in the form of categorization autonomy. We found that self-defined categories can influence important factors such as perceptions of restrictiveness and intention to use. Moreover, we found the nature of the task to be a determining factor in the effect, with the difference being more prominent in higher task flexibility or exploratory searches (i.e., “when”). One explanation for this finding might reside in the fact that creating individualized categories engages one's brain in an exploratory mindset.

Prior research has also studied eye-tracking movements in the two scenarios of exploration and exploitation and found that participants in the exploratory task demonstrated greater amounts of decision-making creativity [55]. Similarly, in our experiment, the greater freedom offered by categorization autonomy could be a reason that task flexibility had a significant moderating effect on search precision. Moreover, our findings contribute to the ongoing refinement of Cognitive Fit Theory within the DSS literature. It has been argued that fit is not merely a function of task-structure alignment but may also reflect adaptive mechanisms and user-level strategies [56]. In line with this, we find that categorization autonomy can support decision effectiveness when users are given the flexibility to tailor information structures to their goals. However, consistent with prior work (e.g., [34]), our results suggest that fit is sensitive to task framing and user involvement.

From practitioners' perspective, this research shows how categorization autonomy can benefit a platform used by general users, such as e-commerce. Categorization autonomy would enable users to find the items they are looking for more accurately. Additionally, by offering categorization autonomy on a website, an organization might improve key consumer behavior factors such as intention to use. While our findings show that categorization autonomy improves both search precision and behavioral intentions, we recognize that these effects may not directly translate into real-world adoption. In practice, factors such as user learning curves, training demands, organizational resistance to change, and integration with existing workflows can all impact adoption outcomes [57]. These issues are beyond the scope of our study, but they represent important considerations for practitioners evaluating the viability of categorization autonomy in live systems.

Our recommendation for websites is to give users the option to choose between self-defined and pre-defined categorization and inform them of the benefits of categorization autonomy. Further, since the openness trait of individuals is positively correlated to the extent to which they may differ in the way they categorize product trees (Table 6, supported by [52]), platforms catering to a more open psychographic profile may benefit more from offering categorization autonomy. However, even as this research aims to demonstrate how categorization autonomy could be beneficial, we acknowledge the utilities that pre-defined categorization might offer. First of all, pre-defined categorization provides cognitive economy. Even if a schema created by the developer might not be fully congruent with the users' mental models, it is still easier to comprehend than viewing a list of all categories in a short period of time. Moreover, creating personalized categorization schemes can be quite time-consuming. For one-time shoppers from a given e-

⁵ <https://www.ynab.com/>

⁶ <https://www.reddit.com/r/ynab/>

commerce website, this may not be a valuable time investment. The benefits of creating personalized categories are best realized for return customers and frequent shoppers on a given platform. We also note that a one-size-fits-all approach is not the best solution. In other words, forcing all users to create their own categories may not have a universally beneficial effect. Similarly, a prior study found that Intel's verbal advisor that increased successful downloads by 27 % was "not for every customer. Less verbal and more analytic customers found the verbal advisor annoying and preferred a more graphic list" [2, p. 202]. Hence, we suggest that both options of full categorization autonomy and predefined schemes be offered and the choice be left to the users.

Lastly, on many e-commerce platforms, shoppers use the search function to find the items they are looking for (e.g., an HDMI cable, or a phone charger). We argue that such a task is more in line with the exploitation search type. In cases where a person is deciding on how to use a gift card they received, for example, they might browse through the product trees, and that exploration could benefit most from categorization autonomy.

5.2. Limitations

Prior research points to the varying levels of exploration requiring different levels of creativity [55]. In our study, the tasks were simplified and thus, did not fully capture the whole spectrum of information requirements. Tasks entailing more rigid frameworks, as well as objectives with greater levels of exploration freedom can be examined in the future.

This research was conducted on an experimental website. While it allowed us to investigate the effects, we realize that real-world settings might yield different results for behavioral intentions - academic projects such as ours may never be able to capture the ramifications of spending actual money on real products. Moreover, in real-world cases, users (of e-commerce or organizational systems) tend to interact with a given system over a continued period of time – as opposed to our study where users spent one hour interacting with the system on average. A longitudinal study could offer additional insights on interactions with a system that offers categorization autonomy. Furthermore, while our use of Cognitive Fit Theory helps explain performance differences in the short term, we acknowledge that cognitive fit may evolve over extended usage. Factors such as learning, adaptation, or user fatigue could alter how well a system aligns with a user's cognitive style over time. We therefore apply CFT specifically in the context of immediate system use and recognize that longer-term cognitive alignment remains an open area for future investigation. We also cannot rule out the possibility that perceived effort or user investment may have contributed to the observed effects on performance of users with full categorization autonomy. Although, the results from the partial autonomy condition could offer support that the improved search precision and higher usage intentions were not only the result of effort and time investment with the interface.

5.3. Future research

There are several points of further research opportunities that we wish to indicate. First and foremost, additional studies are required to understand the deeper cognitive differences between exploratory and exploitive searches when the users create their own categories. It would

be interesting to study the differences in cognitive load by observing brain activity using fMRI and electroencephalography (EEG) tools, or tracking eye movements.

We believe the benefits of categorization autonomy can be realized in contexts other than e-commerce as well. For example, users of banking applications could benefit from control over how expenses are categorized (e.g., 'car insurance' along with 'health insurance' under the category of 'insurances', or grouped next to 'fuel expenses' and 'road taxes' under 'transportation' or 'car ownership' categories). Future research might also involve different product types (e.g., specialized items such as a hydraulic pump for a production facility versus a commodity like a carton of milk) to see if users' prior expertise and knowledge in the subject area could impact their interactions with a system that offers categorization autonomy. Studying the impact of categorization autonomy in high-stakes or expertise-driven domains – where, in certain circumstances, it could even be counterproductive – is certainly an interesting topic for future research. In fact, one could explore the trade-off between user autonomy and system-imposed structure, and how each may enhance or hinder performance, particularly in contexts with varying levels of task flexibility (routine or closed vs. creative or open) and task complexity (low vs. high expertise). Based on task demands, hybrid architectures could also be considered, where user-defined and system-generated categorization approaches are effectively combined in response to user characteristics and task requirements.

Moreover, scholars could examine users' willingness to engage in categorization for subjects with varying degrees of attachment to a system (i.e., new users versus veterans), what the barriers are, and in what factors may contribute to their long-term usage of categorization autonomy.

Our post-hoc analysis shows that the openness personality trait can determine how divergent the individuals' categorization trees are from the baseline. Identifying additional antecedents that may influence users' categorization scheme could be a direction for future research.

Another area for continuing this research program might lie within understanding how individuals create categories. Our work was based on analyzing the 'product' of a user's categorization activity. Future studies on the process itself, by collecting clickstream data for example, could shed more light on the intricacies of the cognitive categorization process.

CRediT authorship contribution statement

Arash Saghafi: Writing – review & editing, Writing – original draft, Validation, Supervision, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Poonacha Medappa:** Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Data curation, Conceptualization. **Ariton Debrliev:** Writing – original draft, Methodology, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Scale items

Table A1

Survey Questions - Adapted from [37].

-
- 1: I had limited control over the way the presented information was categorized.
 - 2: In terms of my preferred way of viewing the information, the categorization was confined.
 - 3: In terms of my preferred way of viewing the information, the categories were restricted.
 - 4: Next time I need to perform such tasks, I would like to use this kind of categorization
 - 5: Assuming I had access to the provided categorization, I would use this website again in the future.
 - 6: Given I had access to the categorization offered by this site, I would use it to perform such tasks if needed.
- QTrap: If you are reading this, please answer 'somewhat disagree'
-

Appendix B. Validity and robustness

Table B1

Validity and Reliability of Constructs.

Item	Intention to use	Restrictiveness
Intention to use 1	0.963	
Intention to use 2	0.972	
Intention to use 3	0.959	
Restrictiveness 1		0.896
Restrictiveness 2		0.900
Restrictiveness 3		0.948

Table B2

Reliability and convergent validity.

	Cronbach's Alpha	Average Variance Extracted
Intention to use	0.962	0.930
Restrictiveness	0.902	0.837

Table B3

Discriminant validity – heterotrait-monotrait ratio.

	Categorization	Intention to use	Restrictiveness
Categorization			
Intention to use	0.333		
Restrictiveness	0.596	0.776	

Appendix C. Scenarios

Information Exploitation Scenario:

Imagine you are an office manager tasked with purchasing the following items:

- Two new monitors for the office workers. The office needs a 24" monitor and a 27" monitor.
- The office microwave is broken! Find a microwave cheaper than \$700 with at least 2-cubic foot of space.
- Please buy a corded Motorola brand phone with cordless handset.
- For calls from your PC you are supposed to buy an on-ear headphone with a visible microphone.
- A cellphone power bank.
- Finally, please find a pencil sharpener.

Information Exploration Scenario:

Suppose you are buying three distinct gifts for your best friend's wedding to satisfy the following three criteria:

You know that one of the newlyweds is a cooking enthusiast, you also know that both of them are very active and enjoy outdoor activities. In addition, you know that they love watching movies at home.

Appendix D. (Behavioral Data)

To extend our understanding of the consequences of categorization autonomy, we developed a set of hypotheses linking autonomy to users' satisfaction, trust, and intentions to use the system. Hypothesis 5 posits that categorization autonomy increases user satisfaction, as the ability to impose one's own organizational logic enhances cognitive alignment and perceived control. Hypothesis 6 proposes that this relationship is mediated

by trust, such that autonomy fosters trust in the system, which in turn drives satisfaction. And, Hypothesis 7 suggests that greater satisfaction leads to higher intentions to use the system, consistent with existing models of IS success and technology adoption. Fig. D-1 illustrates the research model with only the subjective variables (i.e., excluding H1 and H2, namely, Search Precision and Task Flexibility's moderation effect).

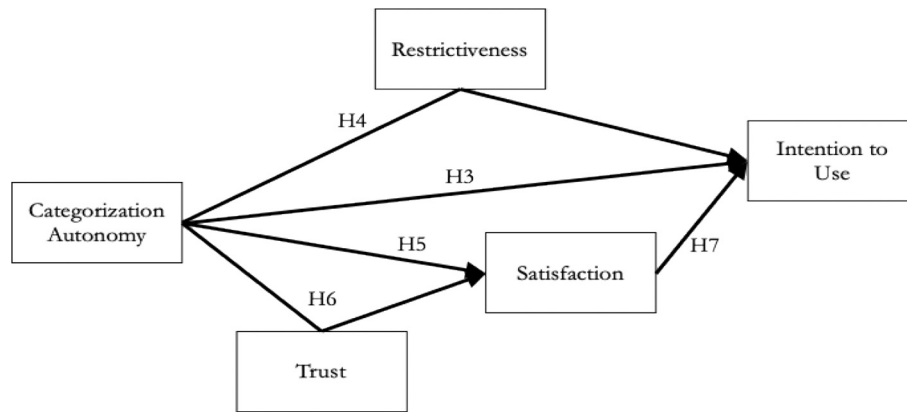


Fig. D-1. Behavioral research model.

The results offer strong support for all three hypotheses. H5 was supported, with categorization autonomy significantly increasing satisfaction ($\beta = 0.323, p = 0.009$). For H6, we found that categorization autonomy significantly increased trust ($\beta = 0.347, p = 0.001$), and that trust strongly predicted satisfaction ($\beta = 0.858, p < 0.001$), fully mediating the direct effect of autonomy (which became non-significant when trust was included). This provides evidence for full mediation, supporting H6. Finally, H7 was also strongly supported, as satisfaction emerged as a robust predictor of usage intentions ($\beta = 0.935, p < 0.001$). Collectively, these findings reinforce the role of categorization autonomy not only in improving task performance, but also in enhancing users' affective and behavioral responses to the system through mechanisms of trust and satisfaction. Table D-1 offers the full statistical analysis using regression.

Table D1
Regression analysis on additional behavioral hypotheses.

Hypothesis	Path	Beta (Unstandardized)	p-value	Supported
H5	Categorization Autonomy - > Satisfaction	0.323	0.009	Yes
H6	Categorization Autonomy - > Trust	0.346	0.001	Yes
	Trust - > Satisfaction (with Autonomy)	0.858	0	Yes
	Categorization Autonomy - > Satisfaction (+ Trust)	0.026	0.775	No (Full Mediation)
H7	Satisfaction - > Usage Intentions	0.935	0	Yes

Data availability

Data will be made available on request.

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